# THE FIBRE CRITERION FOR FLATNESS IN THE NOETHERIAN CASE

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*Allowed standard prerequisites (editorial).*

- The local criterion for flatness. Invoked at the end of the counted proof (line 24239). Converts the vanishing of a Tor group against the residue field into flatness.
- Finitely generated modules over local rings and Nakayama. Used in the counted proof (lines 24214-24236). Propagates flatness from the fibre to the whole module.

*Source reference.* Original lemma 00DV: sources/stacks-algebra/pinned-original-source.tex, source lines 3833–3833. Navigation pointer for \ref{lemma-NAK} used inside the counted span (line 24224); the argument that consumes it is retained in the span.

*Source reference.* Original lemma 00HP: sources/stacks-algebra/pinned-original-source.tex, source lines 9341–9341. Navigation pointer for \ref{lemma-ff} used inside the counted span (line 24226); the argument that consumes it is retained in the span.

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*Date*: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 00MK: sources/stacks-algebra/pinned-original-source.tex, source lines 23800–23800. Navigation pointer for \ref{lemma-local-criterion-flatness} used inside the counted span (line 24239); the argument that consumes it is retained in the span.

*Source reference.* Original lemma 00ML: sources/stacks-algebra/pinned-original-source.tex, source lines 23968–23968. Navigation pointer for \ref{lemma-variant-local-criterion-flatness} used inside the counted span (line 24222); the argument that consumes it is retained in the span.

*Source reference.* Original lemma 00M6: sources/stacks-algebra/pinned-original-source.tex, source lines 18907–18907. Navigation pointer for \ref{remark-Tor-ring-mod-ideal} used inside the counted span (line 24221); the argument that consumes it is retained in the span.

**Lemma 0.1** (Critère de platitude par fibres; Noetherian case)**.** *Let $R$, $S$, $S'$ be Noetherian local rings and let $R \to S \to S'$ be local ring homomorphisms. Let $\mathfrak{m} \subset R$ be the maximal ideal. Let $M$ be an $S'$-module. Assume*

1. *The module $M$ is finite over $S'$.*
2. *The module $M$ is not zero.*
3. *The module $M/\mathfrak{m}M$ is a flat $S/\mathfrak{m}S$-module.*
4. *The module $M$ is a flat $R$-module.*

*Then $S$ is flat over $R$ and $M$ is a flat $S$-module.*

*Proof.* Set $I = \mathfrak{m}S \subset S$. Then we see that $M/IM$ is a flat $S/I$-module because of (3). Since $\mathfrak{m} \otimes_R S' \to I \otimes_S S'$ is surjective we see that also $\mathfrak{m} \otimes_R M \to I \otimes_S M$ is surjective. Consider

$$\mathfrak{m} \otimes_R M \to I \otimes_S M \to M.$$

As $M$ is flat over $R$ the composition is injective and so both arrows are injective. In particular $\mathrm{Tor}_1^S(S/I, M) = 0$ see Remark 00M6. By Lemma 00ML we conclude that $M$ is flat over $S$. Note that since $M/\mathfrak{m}_{S'}M$ is not zero by Nakayama's Lemma 00DV we see that actually $M$ is faithfully flat over $S$ by Lemma 00HP (since it forces $M/\mathfrak{m}_S M \neq 0$).

Consider the exact sequence $0 \to \mathfrak{m} \to R \to \kappa \to 0$. This gives an exact sequence $0 \to \mathrm{Tor}_1^R(\kappa, S) \to \mathfrak{m} \otimes_R S \to I \to 0$. Since $M$ is flat over $S$ this gives an exact sequence $0 \to \mathrm{Tor}_1^R(\kappa, S) \otimes_S M \to \mathfrak{m} \otimes_R M \to I \otimes_S M \to 0$. By the above this implies that $\mathrm{Tor}_1^R(\kappa, S) \otimes_S M = 0$. Since $M$ is faithfully flat over $S$ this implies that $\mathrm{Tor}_1^R(\kappa, S) = 0$ and we conclude that $S$ is flat over $R$ by Lemma 00MK. $\square$